

# AMAN MURARI SINGH

## AI Researcher & Machine Learning Engineer

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### Professional Summary

AI Researcher and Machine Learning Engineer with hands-on experience in Large Language Models, Reinforcement Learning, RAG systems, Multi-Agent AI, and scalable Transformer architectures. Experienced in building AI products, designing efficient LLM variants, implementing distributed training with PyTorch DDP, and developing production-ready AI applications. Open-source work includes nano-GPT-OSS, which reached 100+ GitHub stars and 30+ forks.

### Technical Skills

**Languages:** Python, JavaScript, Rust, PHP, HTML, CSS | **AI/ML:** Deep Learning, NLP, Computer Vision, Reinforcement Learning, Generative AI

**LLMs & GenAI:** Transformers, LLMs, SLMs, RAG, Fine-tuning, RL Fine-tuning, Multi-Agent Systems, Prompt Engineering

**Architectures:** CNN, RNN, LSTM, GAN, Seq2Seq, Attention, MoE, MHA, GQA, MLA | **RL:** PPO, DQN, GRPO, reward design

**Frameworks:** PyTorch, TensorFlow, FastAPI, Django, OpenCV, LangChain, CrewAI, AutoGen | **Tools:** Docker, Git, GitHub, PostgreSQL, MySQL, Pinecone, Weaviate

### Work Experience

#### AI Engineer | Doubt Buddy | Sep 2025 - Present

- Built Datagentz, an AI-powered Microsoft Excel add-in that enables natural-language spreadsheet analysis, automated insights, pattern detection, data transformation, and smart visualization.
- Designed AI workflows for spreadsheet understanding, PDF-to-data extraction, conversational analytics, and Excel-native user interactions for non-technical users.
- Worked across product development, AI feature design, backend integration, user-flow optimization, and marketplace readiness; published on Microsoft Marketplace: marketplace.microsoft.com/en-us/product/WA200009638

#### AI Research Intern | Vizura Technologies Private Limited | May 2025 - Oct 2025

- Contributed to nano-GPT-OSS, an open-source research project for efficient Transformer variants and small-scale language model training.
- Researched and implemented hybrid attention using sliding-window attention and sink tokens, plus MoE-enhanced MLP blocks for improved model capacity and efficiency.
- Achieved 12-30% lower validation loss compared to GPT-2 across 6-12 layer models, with 11% lower RAM usage and 11% smaller disk footprint.
- Demonstrated stronger scaling behavior with a 30.5% validation loss improvement at 12 layers, indicating better generalization at larger depths.
- Implemented multi-GPU distributed training using PyTorch Distributed Data Parallel and open-sourced code, logs, and metrics; repository reached 100+ stars and 30+ forks.

### Selected Projects

#### nanoKimi - Efficient LLM Research Pipeline

- Built a lightweight LLM research pipeline inspired by nanoGPT and Kimi-K2, integrating Muon optimizer, Mixture of Experts, and Latent Attention for compact training experiments.
- Benchmarked against nanoGPT using training speed, inference throughput, model quality, and parameter efficiency; GitHub: github.com/amanmurari/nanokimi

#### Multimodal Retrieval-Augmented Generation System

- Built a multimodal RAG system for text and image retrieval using embedding-based search and FAISS similarity indexing for document search, image search, and question answering.

#### Reinforcement Learning Autonomous Traffic Control System

- Developed an RL-based traffic-control simulation with custom reward functions, state-action logic, traffic scenarios, and feedback-based policy improvement to optimize signal control.

#### Llama2 Vision-Language Model from Scratch

- Implemented core components of a Llama2-style vision-language model, including Transformer blocks, token embeddings, visual feature integration, and text-image reasoning logic.

#### Neural Machine Translation: English to Hindi

- Built an LSTM Seq2Seq neural machine translation system using PyTorch and attention, including preprocessing, tokenization, training, and inference flow.

### Education

Bachelor of Technology, Siksha O Anusandhan University | Sep 2024 - May 2028

### Certifications

TensorFlow Bootcamp | Data Structures and Algorithms in Java - Apna College | Multi-Agent AI Systems Certification

### Achievements

- 1st Place, AI/ML Hackathon - Synchronize 4.0: built a machine learning solution using Python and cloud APIs while competing against 100+ teams.
- Open Source Impact: Core contributor to nano-GPT-OSS, which reached 100+ GitHub stars and 30+ forks; worked on efficient attention, MoE blocks, distributed training, and reproducible evaluation.